BIA 568-B Assignment 2 AI Knowledge Assistant Memo

From: [your name]

Date: October 11, 2026

---

1 Find handbook information

What changed for each of Q01–Q06 when RETRIEVAL_K increased from 1 to 3? Use the retrieved and required passage IDs to explain.

Raising RETRIEVAL_K from 1 to 3 raised required-source recall from 4/7 (57%) to 7/7 (100%) and complete questions from 2/5 to 5/5. Q01 (required S01) and Q05 (required S09) were already correct at K = 1. K = 3 only added unneeded passages (S03, S02 for Q01; S06, S05 for Q05). Q02 (required S04) failed at K = 1: BM25 ranked S11 (supplemental and spouse life) first. Its only match was the generic word "need", while S04 matched only "proof". Keyword search misses that "married" means "marriage" and "HR" means "Human Resources", and the longer S04 passage was penalized for its length. S04 came second. K = 3 recovered it (S11, S04, S01). Q03 (required S07, S06) and Q04 (required S03, S10) each need two topics, so one passage could never be enough. K = 1 returned only S07 (the copay) and only S03 (the start date). K = 3 added S06 (urgent care is not subject to the deductible) and S10 (1x earnings up to $150,000). Q06 has no required passage. At both settings BM25 still returned keyword matches (S14 commuter at K = 1; S14, S10, S08 at K = 3), none of which mention home internet. A larger K therefore gives the model more off-topic text it could misuse. The tradeoff: K = 3 finds every required passage but adds roughly one or two irrelevant passages per question.

2 Compare assistant configurations

Which configurations scored best on Q01–Q05, and which responded fastest? Use the quality and time tables. For Q03 or Q04, explain how the answers changed. For Q06, explain how the assistant handled missing information.

[TO COMPLETE after running Section 2 in Colab — use your own results.]

3 Try the assistant

Give the two questions you tried. Could an employee use each answer? Describe one useful feature or one problem you noticed.

[TO COMPLETE after running Section 3 in Colab — use your own results.]

4 Inspect the workflow

Which workflow step took the longest? Explain how retrieval and the answer model contributed to one of your questions.

[TO COMPLETE after running Section 4 in Colab — use your own results.]

5 Recommend a configuration

Which configuration would you recommend? Support your choice with quality and timing result. Name one improvement before launching the assistant.

[TO COMPLETE after running Section 2–4 in Colab — use your own results.]